

Element I: Testing, Data Collection, and Analysis

Data Summary:

- Thrusters operated within tolerance at all depths
- Buoyancy positive at surface, negative below 3ft (documented flip at 5.7ft)
- Controller lag under 75ms average
- Video review confirmed instability triggered during cable tension spike

Data clearly shows failure cause was inadequate tether lift. Redesign planning now integrates float-linked cable supports.

Charts and logs attached separately document 5-trial averages per test, with standard deviation listed for each.